National University of Computer and Emerging Sciences
Karachi Campus

Database Systems (CS2005)

Mid 1 Exam

Date: Monday, Sep 22nd 2025

Course Instructor(s): Dr. Zulfiqar Ali Memon, Dr. Farrukh Salim, Ms. Atiya Jokhio, Ms. Javeria Farooq, Mr. Basit Ali, Ms. Hajira Ahmed, Mr. Ali Naseer, Mr. Huzaifa, Mr. Umair

Total Time (Hrs): 1
Total Marks: 15
Total Questions: 3

Do not write below this line

Attempt all the questions.

CLO # 1: *Explain fundamental database concepts*

Q1: [marks: 4] [Estimated Time: 10 minutes]

1. GlobalTech, which manages a complex database system for its global operations. GlobalTech uses a sophisticated DBMS that adheres to the three-schema architecture. The system is designed to support various user applications. Describe how the three-schema architecture ensures program-data independence in GlobalTech's system. How does this separation benefit the Sales Application, Finance Application, and IT Management in terms of development and maintenance?

2. Suppose you are at NU University and want to access a database book. Create a diagram showing how a user retrieves data from a database system. Label and describe each component involved and explain the interactions between the user and the database.

CLO # 1: *Explain fundamental database concepts*

Q2: [marks: 2+2+2= 6] [Estimated Time: 25 minutes]

1. Write DDL statements to create relations for Society and members. Carefully read the scenario and apply all possible constraints mentioned below. Don't mention any extra constraint(s) that are not specified in the question.

   a) Societies (SocietyID, SocietyName, Category, Budget)
   i. Each society has a unique SocietyID.
   ii. SocietyName must be unique.
   iii. Category defaults to 'Creativity' if not specified. Category can be (e.g., Technical, Creativity, Sports),
   iv. Budget must be greater than 0.

   b) Members (MemberID, Email, Semester, SocietyID)
   i. Each member has a unique MemberID.
   ii. Email must be unique.
   iii. Semester must be between 1 and 8.
   iv. SocietyID to which they belong, if a society is deleted, all its members should also be deleted.

2. Suppose the university decides that deleting a society should not delete its members but instead set their SocietyID to NULL. Write the modified foreign key constraint.

CLO # 3: Demonstrate an understanding of normalization theory to normalize the database and formulate, using SQL & relational algebra, solutions to a broad range of query & data problems.

Q3: [marks: 5] [Estimated Time: 25 minutes]

A database is designed to store information about cricket matches. The schema is as follows:

- Matches (match_id, match_date, venue_id, team1_id, team2_id, winner_id)
- Teams (team_id, team_name, country)
- Players (player_id, player_name, team_id)
- Match_Scores (match_id, player_id, score)
- Venues (venue_id, venue_name, city)

Write SQL Queries for the following questions:

1. Find the names of all players who scored more than 100 in any single match.
2. List the names of all venues where the 'Pakistan' team has won a match.
3. Fetch the names of all players who have a score of exactly 50 in any match.
4. Print the names of players who have an average score of over 75.
5. List the names of all players who scored more than 100 in a match played at 'Gaddafi Stadium'.

Good Luck!